

IMPACT ANALYSIS

CR-002: CSV Upload and Student Form Auto-Population for the Add New Student Form

1. Change Summary

CR-002 introduces CSV file upload functionality to the administrator's Add New Student form. Administrators will be able to upload a CSV file containing student information, read the CSV data, map supported columns to the corresponding student form fields, and automatically populate the form with the imported values. The administrator can review and edit the populated values before submitting the student record.

2. Impact Assessment

Area	Impact Assessment
Frontend	Moderate - CSV upload control, file handling, parsing feedback, field mapping, form population, validation messages, and UI states will be added to the Add New Student form.
Backend	Low - Existing student creation functionality can remain in use. Backend changes are only required if CSV processing or additional validation is implemented server-side.
API	Low - Existing student creation API remains in use for final form submission. No separate CSV API is required if parsing is performed on the frontend.
Database	None - No database schema change is required because imported values are ultimately submitted through the existing student creation process.
Authentication	None - Existing authentication remains unchanged.
Authorization	None - Existing administrator protection remains unchanged.
Performance	Low - CSV files should be processed locally for the expected student import size. File-size and row-count limits should be applied.
CRUD	Low - Existing Create functionality must continue to use validated form values. Existing Edit/Delete functionality is not expected to change.
Deployment	Low - No database migration is required. Frontend changes and any required validation/configuration changes must be deployed.

3. Frontend Impact

The main affected area is the administrator's Add New Student form. The CSV workflow should be designed as an additional input method without removing the existing manual entry workflow.

- CSV file upload control.
- Accepted file type validation for CSV files.
- CSV parsing and header detection.
- Mapping CSV columns to the existing student form fields.
- Automatic population of the Add New Student form.
- Validation and clear error messages for missing or invalid values.
- Preview/review of imported values before final submission.
- Ability to edit imported values manually before saving.
- Clear/reset CSV import functionality.
- Loading, success, and failure states.
- Responsive styling for supported screen sizes.

The implementation must preserve the existing form structure and manual data-entry option. Uploading a CSV must not automatically create a database record without administrator review and the normal Create/Submit action.

4. CSV Import and Form Mapping Impact

The CSV import process should treat the existing Add New Student form as the source of truth for supported student fields. CSV headers should be matched to the corresponding form fields. The system should handle partial CSV data without incorrectly overwriting unrelated form values.

- Detect and read the CSV header row.
- Match supported CSV column names to the corresponding form fields.
- Populate matching fields with imported values.
- Identify missing required fields before submission.
- Reject or clearly report unsupported or invalid values.
- Allow the administrator to correct imported values manually.
- Prevent duplicate or ambiguous field mapping.
- Keep the final submission path consistent with normal student creation.

5. Backend, API and Database Impact

The preferred implementation is to parse the CSV on the frontend and use the existing student creation API after the administrator reviews the populated form. Under this approach, no new API endpoint is required and the existing backend student-creation validation remains responsible for final record creation.

- Existing student creation API continues to handle final submission.
- No separate CSV upload API is required for client-side parsing.
- No database schema changes are required.
- Existing backend validation must still be applied to imported values.
- Existing duplicate/unique constraints must continue to protect student data.

6. Security Impact

The existing authentication and authorization mechanisms remain unchanged. CSV import must operate only within the administrator's protected Add New Student workflow.

- Do not bypass existing administrator authorization.
- Do not expose imported student information to unauthorized users.
- Validate file type and reasonable file size before processing.
- Treat CSV contents as untrusted input.
- Escape and safely render imported values in the form.
- Do not execute formulas, scripts, or active content from CSV data.
- Continue using existing backend validation and database security controls.

7. CRUD and Functional Impact

The primary CRUD impact is on Create functionality. CSV import must only prepare the Add New Student form; the normal Create operation must remain responsible for saving the student record.

- Manual student creation must continue to work without CSV upload.
- CSV-populated student data must be editable before submission.
- Create must use the final reviewed form values.
- Existing Edit and Delete operations must remain unaffected.
- Validation errors must identify the affected field or imported value where possible.

8. Performance Impact

The expected performance impact is low for normal student CSV files because parsing can be performed locally in the browser. The implementation should enforce a reasonable file-size and row-count limit to avoid excessive browser memory usage. Large-scale bulk import is outside the current scope unless a future change introduces server-side processing.

9. Testing Impact

- Upload a valid CSV file.
- Reject a non-CSV or unsupported file.
- Detect the CSV header row correctly.
- Map supported columns to the correct student form fields.
- Populate all supported fields correctly.
- Handle partial CSV data.
- Handle missing required values.

- Handle invalid field values.
- Handle empty CSV files.
- Handle CSV files with only headers.
- Handle extra or unsupported columns.
- Handle leading and trailing spaces in imported values.
- Handle different letter casing in supported headers where applicable.
- Review and manually edit imported values.
- Clear or reset imported values.
- Submit the populated form successfully.
- Verify the created student record contains the final reviewed values.
- Verify existing manual student creation still works.
- Verify administrator authentication and authorization.
- Verify responsive behavior on supported screen sizes.

10. Risk Analysis

Risk	Impact	Mitigation
Incorrect CSV-to-form field mapping	High	Define supported headers clearly, validate mappings, and test each mapped field.
Invalid or incomplete imported student data	High	Run the same required-field and value validation used by normal form submission.
Unexpected CSV content	High	Treat CSV values as untrusted input and safely handle and render all imported data.
Duplicate student records	High	Preserve existing backend uniqueness checks and duplicate validation.
CSV parsing failure	Medium	Show a clear import error and allow the administrator to continue with manual entry.
Large CSV affects browser performance	Medium	Apply file-size and row limits and keep large-scale import outside the current scope.
Existing Create functionality breaks	High	Perform regression testing of the existing Add New Student workflow.
Imported values are submitted without review	High	CSV upload only populates the form; final creation requires the existing Submit/Create action.
Responsive UI problems	Low	Test upload and form states across supported screen sizes.

11. Overall Impact and Recommendation

Overall Impact: Moderate

CR-002 is primarily a frontend enhancement to the administrator's Add New Student workflow. It introduces CSV file handling, data parsing, field mapping, validation feedback, and form auto-population while preserving the existing manual entry and student creation process.

No database schema change is expected. No new backend API endpoint is required if CSV parsing is implemented on the frontend. The existing authentication, authorization, and final student creation controls should remain unchanged.

The change is technically feasible and recommended for implementation, subject to formal approval, CSV format and field-mapping definition, security validation, and subsequent functional and regression testing.

Key implementation consideration: CSV upload must populate the existing Add New Student form rather than directly creating records. The administrator must be able to review and correct the imported values before using the existing Create/Submit action.

Reference basis: CR-001 Impact Analysis structure and the Student Management System CHANGELOG Version 1.1.0 conventions. CR-002 is the proposed change described in this document.